

Mathematical Foundations for the Upgraded Chaos Language Algorithm: Coding-Theoretic Semantics, Separation Theorems, Causal-State Categories, and the Corrected Quantale Bridge

Working notes for Ben Goertzel, prepared with Claude (Anthropic)
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Abstract

These notes supply the mathematical backbone for the July 2026 upgrade of the Chaos Language Algorithm (CLA) and its `chaoslang` implementation. Part I gives the new adaptive/prequential MDL objective an exact coding-theoretic semantics, including the implicit prior of the one-part inline code. Part II proves where code-length differences can come from at all (the Invariance Lemma), derives a recurrence-time ceiling showing that exact-repeat chunk grammars are confined to the Lempel–Ziv redundancy class on positive-entropy sources, and establishes a separation theorem locating the genuine home turf of grammar codes: hierarchically (substitution-)generated sources, where grammar codes beat every Markov, context-tree, and LZ control by polylogarithmic versus nearly-linear total bits. Part III develops the automata track: the ε -machine is the MDL-optimal grammar of a sofic source, MDL category acceptance is a consistent test of contextual exchangeability, and—combining the two—CLA meta-symbols on sofic sources are exactly the fibers of the alphabet’s action on causal states, i.e. kernel classes of the syntactic-monoid map. Part IV corrects and sharpens the McBride-derivative/quantale bridge: in the additive quantale the McBride derivative of a proposal is the exact MDL first difference, emergent synergy is the mixed second difference (supermodular edit interaction), and a closed-form synergy estimator yields a principled composite-proposal generator. Two new falsifiable predictions follow: chunk-grammar CLA should win decisively on substitution-structured dynamical regimes (period-doubling accumulation points, Sturmian codings) and lose to causal-state coding on fully hyperbolic chaos—reframing CLA as a detector of hierarchical generativity rather than of chaos per se. Standard results are cited as [std]; project-specific claims carry proof sketches.

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1 Setup and notation

Let $\mathcal{S} = (S_t)_{t \geq 1}$ be a stationary ergodic process over a finite alphabet V , $|V| = m$, with entropy rate $h = \lim_n \frac{1}{n} H(S_{1:n})$. Write $h_k = H(S_{k+1} \mid S_{1:k})$ for the order- k conditional entropy, so $h_k \downarrow h$.

The source classes of interest are nested as

$$\text{IID} \subset \text{Markov-}k \subset \text{UHMM (unifilar hidden Markov / sofic)} \subset \text{stationary ergodic},$$

together with a separate, partially overlapping class SUB of *substitution/hierarchically generated* processes: fixed points of morphisms, linearly recurrent words, and stochastic context-free outputs. Deterministic members of SUB have $h = 0$ yet nontrivial structure at every finite order.

A *code* is a prefix-free map from strings to bits. For a code C and true source P , the expected *redundancy* is $\rho_n(C) = \mathbb{E} \ell_C(S_{1:n}) - H(S_{1:n})$. All coders in **chaoslang/coding/** are prequential: $\ell(s_{1:n}) = \sum_t -\log_2 q_t(s_t \mid s_{1:t-1})$ for a predictive system q .

Standard redundancy facts used throughout [std: Krichevsky–Trofimov 1981; Rissanen 1984; Willems–Shtarkov–Tjalkens 1995; Ziv–Lempel; Plotnik–Weinberger–Ziv]:

Coder	Rate-optimal class	Total redundancy ρ_n
KT (adaptive unigram)	IID	$\frac{m-1}{2} \log_2 n + O(1)$
Adaptive Markov- k	Markov- k	$\frac{m^k(m-1)}{2} \log_2 n + O(1)$
CTW, depth D	tree sources of depth $\leq D$	$\frac{k^*(m-1)}{2} \log_2 n + O(\text{tree})$, $k^* =$ true number of contexts
CausalStateCoder, states (two-part)	s UHMM	$L_{\text{machine}} + \sum_{\sigma} \frac{ A_{\sigma} -1}{2} \log_2 n + O(1)$, $A_{\sigma} =$ allowed successors of state σ
LZ78	stationary ergodic (universal)	$O(n \frac{\log \log n}{\log n})$

The decisive scale separation: **parametric coders pay $\Theta(\log n)$ total redundancy on their class; universal dictionary coders pay $\Theta(n/\log n)$; static coders pay $\Theta(n)$ off their class.** The July 2026 experimental programme, in these terms, repeatedly matched a statically coded grammar arm against $\Theta(\log n)$ - and $\Theta(n/\log n)$ -redundancy opponents.

2 The Invariance Lemma: where can bits be saved?

Chunk rewriting is invertible: given the grammar, the rewritten stream determines $S_{1:n}$ and conversely. The same holds for category rewriting with recorded members $M[v]$, and for the factorization of a category-bearing stream into (class stream, member choices) via the chain rule.

Lemma 1 (Invariance). *Let T_G be any invertible grammar-rewriting transformation. Then $H(T_G(S_{1:n})) = H(S_{1:n})$. Consequently, for any lossless coding scheme the expected total code length of a grammar arm satisfies $\mathbb{E}\ell \geq H(S_{1:n})$, and the difference between two lossless arms (grammar versus no-grammar, both using ideal-bits prequential coders) is a difference of redundancies plus model/definition costs only:*

$$\mathbb{E}\Delta\ell = \Delta\rho_n + \Delta L_{\text{model}}.$$

Proof. Entropy is invariant under bijections; the lower bound is the noiseless coding theorem; subtract. $\square$

Remark. *No grammar edit can beat the entropy of the data; a grammar “helps” exactly insofar as the rewritten stream is cheaper for the frozen base coder to learn than the raw stream, i.e. insofar as rewriting reduces the effective parametric complexity (number and size of contexts, memory length) of the best predictive model, or shortens the learning transient. This single lemma explains three of the central July findings.*

1. **The 65,536-symbol attractor failures.** *Microstate streams of hyperbolic-like flows are close to Markov (Sections 7–9): the raw stream is already in the base coder’s low-redundancy class, so a redundancy reduction by rewriting is unavailable and any positive model cost loses. More data can never fix this.*
2. **The Fibonacci/paperfolding paradox** (suffix code improved, total lost): *rewriting by substitution rules genuinely shortens the effective memory (large $-\Delta\rho_n$), but the July two-part model code charged parameters at static rates, swamping the gain. Under adaptive member/parameter codes this charge collapses to the $\frac{k}{2}\log n$ scale (Section 4); this is the mechanistic content of prediction P_4 of the evaluation programme.*
3. **The objective is not “compression” naively.** *Two arms with identical asymptotic rates can differ by $\Theta(\log n)$ or $\Theta(n/\log n)$ redundancy; that, not the rate, is the entire battlefield on stationary sources.*

3 What chunking can buy: the recurrence-time ceiling

Proposition 1 (Chunk-length ceiling). *Let $h > 0$. By the Ornstein–Weiss return-time theorem [std], the first return time R_ℓ of an ℓ -block satisfies $\frac{1}{\ell}\log_2 R_\ell \rightarrow h$ almost surely. Hence in a*

stream of length n , blocks of length $\ell \gg \frac{1}{h} \log_2 n$ typically occur once; every chunk rule (which requires at least two occurrences) has body length $O(h^{-1} \log n)$, and the grammar’s dictionary behaves like an LZ-class parse.

Corollary 1 (Chunk grammars are confined to the LZ redundancy class). *For stationary ergodic $\mathcal{S}$ with $h > 0$, any exact-repeat chunk grammar code (under any search) has $\rho_n = \Omega(n/\log n)$ in the worst case over the class, matching LZ up to constants; whereas if $\mathcal{S}$ is additionally UHMM with s states, the CausalStateCoder attains $\rho_n = O(\log n)$.*

Proof sketch. The grammar code’s data part is a parsing into dictionary phrases of length $O(\log n)$ by Proposition 1; phrase-counting converse bounds for dictionary codes on ergodic sources give the $\Omega(n/\log n)$ redundancy floor [std]. The UHMM upper bound is the CausalStateCoder row of the table in Section 1. $\square$

This is the formal version of “chaos gives statistical, not verbatim, regularity,” and it licenses the partial-outcome clause of experiment E7: on hyperbolic-regime fixtures, chunk-CLA cannot beat CSSR/CTW except by constants; a loss there is a theorem-level expectation, not an implementation bug.

Proposition 2 (Separation theorem: the home turf of grammar codes). *Let $\mathcal{S} \in \text{SUB}$ be a linearly recurrent substitution word (e.g. the Fibonacci word, factor complexity $p(k) = k + 1$). Then:*

- (i) *it admits a straight-line program (SLP) / grammar of size $O(\log n)$, so the inline grammar code of Section 5 achieves total length $O(\log^2 n)$;*
- (ii) *every order- k Markov coder pays total at least cn/k bits: there is a right-special factor of every length, whose frequency under the unique ergodic measure is $\Theta(1/k)$, contributing $h_k = \Theta(1/k)$; optimizing k against the $m^k \log n$ parameter cost leaves total cost $\tilde{\Omega}(n/\log n)$;*
- (iii) *LZ78 pays $\Theta(n/\log n)$ (phrase counting on zero-entropy sequences [std]).*

Hence grammar codes separate from all Markov/CTW/LZ controls by $\text{polylog}(n)$ versus $\tilde{\Theta}(n/\log n)$ exactly on hierarchically generated sources.

Remark (Scientific positioning). *The algorithm’s discriminating power is not “chaos detection” but **detection of hierarchical generativity**: CLA should win precisely on substitution-structured dynamics and lose (to parametric automata coders) on fully developed hyperbolic chaos. Section 9 turns this into two new falsifiable fixtures.*

Proposition 3 (Hardness, and an initializer with a guarantee). *The smallest-grammar problem is NP-hard and APX-hard [std: Charikar, Lehman, Liu, Panigrahy, Prabhakaran, Sahai, Shelat 2005]; greedy and Re-Pair have super-constant worst-case approximation ratios, while an LZ77-factorization-based balanced-SLP construction achieves approximation ratio $O(\log(n/g^*))$ to the smallest grammar size g^* [std: Rytter 2003; Charikar et al.].*

Remark (Engineering consequence). *Add **LZ77SLPInitializer** alongside **RePairInitializer** in milestone M-D: it is the only initializer with a worst-case guarantee, and on SUB-class fixtures the guarantee is meaningful ($g^* = O(\log n)$). It also calibrates expectations for experiment E4: search approximates an NP-hard optimum, so the 80%-gap-closure gate is the right relative criterion.*

4 Exact semantics of the adaptive two-part scorer

Fix a grammar structure G (rules, categories, topology—no probabilities). The scorer computes

$$L(G, S) = L_{\text{model}}(G) + \ell_{\text{preq}}(\text{parse}(S) \mid G),$$

where every KT table is a Bayes mixture: for a context with counts $n_1, \dots, n_m$, the prequential KT bits equal $-\log_2 \int \prod_i \theta_i^{n_i} d\pi_J(\theta)$ with π_J the Dirichlet($\frac{1}{2}$) (Jeffreys) prior [std].

Proposition 4 (Bayesian reading). *The adaptive two-part score is exactly $-\log_2 [w(G) m_G(S)]$, where $w(G) = 2^{-L_{\text{model}}(G)}$ is a prefix-code prior over grammar structures and m_G is the Bayes marginal likelihood of the data under G with independent Jeffreys priors on every conditional distribution (base-coder contexts and category member distributions). Minimizing L is MAP structure selection under this prior; the $\frac{k}{2} \log n$ parametric charges are levied implicitly and correctly by the mixture, never by hand.*

This is the precise sense in which the July accounting failure (F3) was structurally unfixable by better serialization alone: static member/parameter tables charged $\Theta(1)$ bits per parameter per *use*; the mixture charges $\frac{1}{2} \log n$ per parameter *total*.

Proposition 5 (Asymptotic soundness of acceptance: no false structure). *Suppose the raw stream is generated by a source in the base coder’s class (e.g. Markov- k with coder order at least k), and let q be any fixed nontrivial edit. Then by Lemma 1 the data-bit change of q is a redundancy difference; rewriting cannot reduce the already-minimal-for-the-class context count, so $\mathbb{E} \Delta L(q) = \Delta \rho_n + L_{\text{model}}(q) \geq L_{\text{model}}(q) - O(1) > 0$ for large n . Hence the probability that MDL accepts q tends to 0. Shuffled-null non-promotion is the finite-sample face of this proposition, now backed by an argument rather than only a control.*

5 The inline one-part code: prefix validity, implicit prior, equivalence

Reserve in every KT table an escape symbol $\uparrow$. A codeword is a sequence of decisions; at each step the decoder holds the same adaptive state as the encoder. Definitions are self-delimiting: $\uparrow$, then an Elias/ $\log^*$ length, then body tokens under the current adaptive coder, in topological order.

Proposition 6 (Decodability). *The inline code is a prefix code on pairs (grammar, parse), hence on the original strings $S_{1:n}$, and the decoder of the specification (Section 1.7) inverts it exactly. This is what the round-trip property test certifies empirically.*

Proposition 7 (Implicit prior). *The inline code realizes the Bayesian score of Proposition 4 with the structure prior replaced by the sequential prior*

$$w_{\text{inl}}(G) = \prod_{r \in \text{defs}(G)} P_{\uparrow}(t_r) 2^{-\log^* |\beta_r|} \prod_j q_{t_r}(\beta_{r,j}),$$

where t_r is the first-use position of definition r , $P_{\uparrow}$ the escape probability there, and q_t the coder’s predictive distribution at that time. In words: **rules that are natural continuations of what the coder has already learned are cheap to declare**. This is a data-dependent Occam prior; it automatically implements the “source-adaptive universal prior” requested by failure item F3.

Proposition 8 (Two-part/one-part equivalence up to escapes). *For any fixed finite grammar G_0 ,*

$$\ell_{\text{inl}}(S) \leq \ell_{\text{preq}}(\text{parse}(S) \mid G_0) + \underbrace{\sum_r \left(\log^* |\beta_r| + \sum_j -\log_2 q(\beta_{r,j}) \right)}_{=O(\text{total definition length} \cdot \log m)} + \underbrace{\Delta_{\uparrow}}_{\text{escape overhead}},$$

with $\Delta_{\uparrow} \leq (\# \text{active tables}) \cdot O(\log n_{\text{table}})$: *reserving one extra symbol in a KT table over m' symbols inflates the table's total by at most $\log_2 \frac{n+\frac{1}{2}(m'+1)}{n+\frac{1}{2}m'}$ per push, summing to $O(\log n)$ per table. Hence the two regimes agree to within additive $\tilde{O}(\text{def. length} \cdot \log m + \# \text{tables} \cdot \log n)$, and in particular have identical rates and identical Θ -class redundancies.*

Remark (Practical corollary). *Run the two-part adaptive scorer (v1) as the workhorse; the prequential scorer (v2) is a consistency check and the cleaner object for theory. Disagreement between them beyond the Proposition 8 slack is a bug detector.*

6 Category edits as parameter tying: a consistency theorem

Fix a frame occurrence set for candidate $M = \{v_1, \dots, v_r\}$. Say the members are *contextually exchangeable* if for all pasts p and futures f occurring with the frame,

$$P(f \mid p, \text{member} = v_i) = P(f \mid p, \text{member} = v_j) \quad \forall i, j. \quad (\text{CE})$$

The category-bearing code factors each occurrence into (class token M , member choice), coding the class token in the base coder's stream and the member with a per-category adaptive KT.

Theorem 1 (Consistency of category acceptance). *Under the adaptive scorer, with the frame occurring $\nu_n \rightarrow \infty$ times:*

- (a) *If (CE) holds, the category edit ties the r per-member successor distributions into one, reducing the free-parameter count by $\Delta k = (r-1) \cdot (\text{successor df})$ at the price of the member code $\approx \nu_n H(\text{member}) + \frac{r-1}{2} \log \nu_n$. Since the member entropy was already being paid inside the untied successor distributions (chain rule), the net change is $\Delta L = -\frac{\Delta k}{2} \log n + \frac{r-1}{2} \log \nu_n + O(1) < 0$ eventually whenever the tying removes at least r degrees of freedom, and the edit is accepted a.s. for large n .*
- (b) *If (CE) fails with average conditional divergence $\bar{D} > 0$, tying incurs data cost at least $\nu_n \bar{D}(1 + o(1))$, which dominates every log-scale saving; the edit is rejected a.s.*

Proof sketch. Both directions are the standard $\frac{k}{2} \log n$ versus nD dichotomy of parametric-mixture code lengths [std], applied to the tied versus untied families; the only CLA-specific point is the chain-rule bookkeeping showing member-choice entropy is not double-charged. $\square$

Remark. MDL category acceptance is a consistent statistical test of contextual exchangeability. *This upgrades the Jensen–Shannon threshold heuristic (a fixed θ on context divergence, neither necessary nor sufficient) to a calibrated test, and it says exactly which planted structures experiments E3/E6 must recover: those where (CE) holds by construction.*

7 The ε -machine: MDL-optimality of the causal quotient

Histories x, x' are *predictively equivalent* if $P(\text{future} \mid x) = P(\text{future} \mid x')$; causal states are the equivalence classes; the ε -machine is the induced unifilar, minimal, sufficient presentation [std: Crutchfield–Young 1989; Shalizi–Crutchfield 2001]. CSSR consistently estimates it for finite-state sources [std: Shalizi–Klinkner 2004].

Theorem 2 (The ε -machine is the MDL grammar of a sofic source). *Among all unifilar finite-state presentations $\mathcal{M}$ of a UHMM source, the two-part adaptive code length satisfies*

$$\ell(\mathcal{M}) = L_{\text{machine}}(\mathcal{M}) + nh + \sum_{\sigma \in \mathcal{M}} \frac{|A_\sigma| - 1}{2} \log_2 n + O(1)$$

for predictively sufficient $\mathcal{M}$, and $\ell(\mathcal{M}) \geq n(h + D_{\mathcal{M}}) + O(\log n)$ with $D_{\mathcal{M}} > 0$ for insufficient $\mathcal{M}$. Every sufficient unifilar presentation refines the ε -machine (causal states are the coarsest sufficient statistic), and refinement cannot decrease $\sum_{\sigma} (|A_\sigma| - 1)$: splitting σ into nonempty σ_1, σ_2 with successor sets covering A_σ gives $(|A_{\sigma_1}| - 1) + (|A_{\sigma_2}| - 1) \geq |A_\sigma| - 1$. Hence the ε -machine minimizes the code length up to $O(1)$, and MDL model selection over unifilar machines recovers it asymptotically.

This theorem is the license for the automata track: on sofic symbolic dynamics—which includes the symbolic dynamics of uniformly hyperbolic systems with Markov partitions [std: Bowen]—the *correct* output of a “chaos language” analysis is a finite automaton with transition probabilities, and the CausalStateCoder is its MDL-optimal code. Chunk/category grammars are, on this class, at best a re-notation of the same automaton and, by Corollary 1, a worse-coded one.

8 Categories as syntactic-monoid fibers

Let $\mathcal{M}$ be the ε -machine with state set Σ and (partial) unifilar transition $\delta : \Sigma \times V \rightarrow \Sigma$. Each token v induces a partial transformation $\delta_v : \Sigma \rightarrow \Sigma$; the monoid generated by $\{\delta_v\}$ is the *transition monoid*, which for the minimal presentation of a sofic shift is the *syntactic monoid* of its language [std].

Definition 1 (Alphabet congruence). $v \equiv w \iff \delta_v = \delta_w$ on all reachable states (equal as partial maps: same domain of definedness, same images).

Theorem 3 (Category–congruence correspondence).

- (a) $\equiv$ is the coarsest equivalence on V such that the quotient process—replace each s_t by its class $[s_t]$ —remains unifilar over the class alphabet with the same state set; the member choice at each step, conditioned on the current causal state and class, is then predictively irrelevant. Thus classes of $\equiv$ satisfy contextual exchangeability (CE) uniformly in every context.
- (b) Conversely, any CLA category all of whose occurrences satisfy (CE) in every frame is a refinement of an $\equiv$ -class on the states where it occurs.
- (c) Therefore, on a sofic source, the categories that Theorem 1 accepts in the large-sample limit are exactly (unions of occurrences of) the fibers of the map $V \rightarrow \text{syntactic monoid}$, $v \mapsto \delta_v$ —which is precisely the rule implemented by *CSSRCategoryInducer* (specification Section 5.3, with the finite-support caveat).

Proof sketch. (a) $\delta_v = \delta_w$ makes the class-quotient dynamics well defined and state-preserving; predictive distributions depend on the history only through the causal state, which the class sequence still determines; hence member choice adds no predictive information. (b) If (CE) holds in every context, then v, w drive every reachable state to predictively equivalent—hence equal—causal states. (c) combines (a) and (b) with Theorem 1. $\square$

Remark (Interpretation). *The paradigmatic pole of CLA—substitutability classes—is, on the automata side, an algebraic object: **meta-symbols are the kernel classes of the alphabet’s action on causal states**. This gives the inducer a correctness criterion (the E6 adjusted-mutual-information gate becomes “does the empirical kernel match the planted kernel”), and it gives the Hyperon migration a clean target: the transition monoid and its kernel are finite relational structures suited to Atomspace representation and MeTTa querying, with PLN-style uncertainty over $\delta_v = \delta_w$ assertions under finite support.*

Remark (Soft categories). *Relax $\equiv$ to statewise partial agreement ($\delta_v = \delta_w$ on a high-probability subset of states, with disagreement mass ϵ); Theorem 1(b)’s penalty then reads $\nu_n \epsilon \bar{D}$, so the inducer can rank soft candidates by estimated $\epsilon \bar{D}$ against the $\frac{\Delta k}{2} \log n$ budget—a principled version of the soft/overlapping categories of repair item R4.*

9 Where chaotic dynamics sits: two falsifiable predictions

Combining Sections 3 and 7:

- **Hyperbolic / fully developed chaos** (Lorenz-like flows away from bifurcation accumulation; Markov-partitioned maps): symbolic dynamics is sofic, or well approximated by sofic at the microstate resolution, with $h > 0$. Predicted ordering of complete codes: CausalStateCoder $\approx$ CTW $\geq$ adaptive-Markov- $k >$ LZ78 $>$ chunk-CLA, with CLA’s deficit explained by Corollary 1. A chunk grammar “detecting chaos” here would *contradict* the theory; experiments E5/E7 test the ordering.
- **Substitution-structured dynamical regimes**: period-doubling accumulation points (the kneading sequence at the Feigenbaum point is generated by a substitution, with $h = 0$), quasiperiodic mode locking (Sturmian symbolic dynamics), and intermittency ladders. These are SUB-class—exactly the separation regime of Proposition 2. Predicted ordering: chunk-CLA with hierarchical rules beats every Markov/CTW/LZ/CSSR control, by polylog versus $\tilde{\Theta}(n/\log n)$.

New preregistrable predictions (to append to the evaluation programme as E5b, with fresh fixtures):

- (P5) On the logistic map at the Feigenbaum accumulation point and on a Sturmian rotation coding, the upgraded CLA (Re-Pair or LZ77-SLP initialization, inline scorer) beats all controls including CSSR, with the margin growing like $n/\log n$ across lengths $\{2^{12}, 2^{14}, 2^{16}\}$.
- (P6) On fresh hyperbolic-regime fixtures, the CausalStateCoder beats chunk-CLA at every length, and the recovered machine’s state count stabilizes across seeds while the chunk grammar’s rule count grows like the LZ phrase count $\Theta(n/\log n)$.

A joint (P5 $\wedge$ P6) outcome would establish CLA’s identity crisply: **a detector of hierarchical generativity in dynamics, complementary to—not competitive with—automata reconstruction of hyperbolic chaos**. For OmegaSim this reframes A6/A7: the interesting question becomes which regime a multi-agent appraisal system’s traces occupy, and transitions between the two regimes (for example at coupling thresholds) become the natural observable.

10 The quantale/weakness bridge, corrected and sharpened

Work in the additive quantale $\mathcal{Q} = ([0, \infty], \otimes = +, \oplus = \inf, e = 0)$ with residuation $a \backslash c = \max(c - a, 0)$. Define the *weakness of a model structure* as its stochastic complexity: $w(G) = L(G, S)$ under the adaptive scorer, equivalently $-\log_2$ of the Bayes score of Proposition 4. This replaces the July bridge note’s mixed conventions with a single consistent instantiation; Definitions 1–2 and Propositions 1–2 of that note survive verbatim under this reading, with the corrections below.

10.1 Edits as one-hole contexts (now exact)

A proposal q is a one-hole context in the grammar–stream pair; its McBride derivative is the exact first difference $\partial_q L = L(G + q, S) - L(G, S) = \Delta L(q)$, computable (not merely estimable) by the scorer’s **delta**. Greedy CLA is steepest descent in this discrete gradient field—literally rather than approximately, because the adaptive scorer’s Δ has no unaccounted parameter terms.

10.2 Synergy is the mixed second difference

For proposals q_1, q_2 define

$$\Delta^2 L(q_1, q_2) = L(G + q_1 + q_2) - L(G + q_1) - L(G + q_2) + L(G).$$

The bridge note’s emergent synergy is $\sigma(q_1, q_2) \geq e \iff \Delta^2 L(q_1, q_2) < 0$ (strict super-additivity of improvements): **emergence equals supermodular interaction of edits on the edit lattice**. Theorem 1 of the bridge note (the $a\{x, y\}b$ construction) is the canonical instance: with the adaptive scorer one computes, for ν frame occurrences,

$$\begin{aligned} \Delta L(q_{\text{cat}}) &\approx +O(\log n), & \Delta L(q_{\text{chunk}}^{(axb)}) &\approx +O(\log n), \\ \Delta L(q_{\text{cat}} + q_{\text{chunk}}) &\approx -\frac{\Delta k}{2} \log n + O(\log \nu) < 0, \end{aligned}$$

so $\Delta^2 L < 0$ with magnitude $\Theta(\log n)$: synergy is real but logarithmic-scale on finite-state examples—consistent with Proposition 2’s message that the linear-scale wins live on SUB-class sources, where the composite move unlocks a hierarchy and $\Delta^2 L = -\Theta(n/\text{polylog})$.

10.3 Answer to Open Question 5: synergy-guided proposals

$\Delta^2 L$ can be estimated *without rescoring* from sufficient statistics: for a category candidate C (of size r) and the frame chunks it would unify, the dominant terms are

$$\widehat{\Delta^2 L}(C, \text{frame}) \approx -\frac{(r-1)d_{\text{succ}}}{2} \log_2 n + \nu \widehat{D}_{JS}(C) + \frac{r-1}{2} \log_2 \nu + c_{\text{def}},$$

with $\widehat{D}_{JS}$ the plug-in mean successor divergence among members and c_{def} the inline definition estimate from Proposition 7. The **CategoryThenChunkProposer** should rank composites by this

statistic, with acceptance still adjudicated by the exact scorer (project rule: no private objectives); the ledger’s estimated-versus-realized $\Delta^2 L$ residuals become a free calibration diagnostic. This is the McBride derivative acquiring the direct algorithmic role the bridge note asked for—as a ranking device.

10.4 Correction to the bridge note’s Proposition 3 (the ODE relaxation)

The claim that the grammar trajectory satisfies a quantale exponential-growth ODE requires $\Delta_{q(t)} \approx \kappa \otimes m(t)$, i.e. improvement proportional to current weakness, which the acceptance rule does not guarantee and which ledger data can falsify. The defensible replacements: (i) $L(G(t), S)$ is nonincreasing and bounded below by $H(S_{1:n})$ (Lemma 1), so the trajectory converges; (ii) the edit lattice with $\Delta^2 L$ of mixed sign is neither sub- nor supermodular globally, which is precisely why greedy fails and why beam/composite search is needed. The correct dynamical statement about CLA’s own search is “gradient descent on a non-modular discrete landscape,” and the interesting “chaotic grammar dynamics” question (bridge note Open Question 4) becomes whether beam-search trajectories on SUB-class inputs exhibit sensitive dependence on tie-breaking—which the deterministic-seed ledgers measure directly.

10.5 Causal quotients as weakness minimization

Theorem 2 in this language: among predicates m_π induced by partitions π of histories, predictive sufficiency is the pattern condition $I(m_\pi) \geq e$ (zero residual divergence), and the ε -machine is the *coarsest* π satisfying it—the weakness-minimizing quotient, since coarsening reduces the $\log n$ -scale weakness term while sufficiency pins the linear term at nh . Theorem 3 then says the alphabet congruence is the restriction of the same universal construction from histories to letters. The whole upgraded system is one principle instantiated three times: **minimize stochastic-complexity weakness; chunks descend it syntagmatically, categories descend it paradigmatically by tying, causal states descend it by quotienting—and the quantale calculus is the common bookkeeping.**

11 Open problems

Open Problem 1 (Stochastic-CFG analogue of Theorem 2). *Is there an MDL-consistency theorem selecting a minimal PCFG among sufficient hierarchical presentations of a SUB-class source, playing the role the ε -machine plays for UHMM? Natural conjecture: yes, with the inline prior of Proposition 7; the obstruction is non-identifiability of PCFGs.*

Open Problem 2 (Rate of Theorem 1). *Finite-sample thresholds: how large must ν_n be, as a function of $\bar{D}$ and r , for the correct accept/reject decision with high probability? This would let experiments E3/E6 predeclare crossover horizons instead of estimating them.*

Open Problem 3 (Approximation ratio of composite-beam search). *On SUB-class inputs, does synergy-guided composite/beam search provably improve on the Rytter $O(\log(n/g^*))$ baseline relative to the smallest grammar, or only empirically?*

Open Problem 4 (Two-regime front-end classifier for OmegaSim). *Find a statistic (e.g. growth exponent of the chunk dictionary versus n , contrasted with causal-state count stability) that classifies a trace as sofic-regime versus substitution-regime before committing to a model class, turning the Section 9 dichotomy into a front-end detector.*

Open Problem 5 (Quantale choice, partially resolved). *In the additive quantale the theory above is exact; the probability quantale $([0, 1], \times, \max)$ corresponds to replacing $\oplus$ -aggregation by best-case (Viterbi-like) scoring, which breaks the mixture semantics of Proposition 4. So the additive quantale is not merely convenient but forced by the Bayesian reading; the remaining question is whether any non-additive quantale supports a coherent robust (minimax-regret / NML) variant.*

Provenance

Companion documents: `chaoslang_upgrade_spec.md` (implementation specification, milestones M-A..M-F), `chaoslang_eval_programme.md` (experiments E0–E8), *CLA Experiments, Results, Failure Diagnosis, and Repair Programme* (2026-07-23), *Bridging the CLA and McBride-Derivative Pattern Calculus* (2026-07-05), and *The Chaos Language Algorithm: An Implementation-Oriented Specification* (2026-07-03). Repository: <https://github.com/bgoertzel-sing/chaos-language-algorithm>.